Polynomial-Linked Tucker Rank Lifts for Nonlinear Tensor Observations

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Abstract

Classical Tucker models explain multiway data with a multilinear low-rank structure. This is effective when the observed tensor is close to a multilinear signal, but many sensing and imaging pipelines include nonlinear entry-level effects, such as response curves, saturation, and material-dependent distortions. These effects can make a low-rank latent tensor appear higher-rank in the observed domain.

We propose Nonlinear Tucker Decomposition with Polynomial Links (NTD-PL), a parameter-tied Tucker enhancement and shared-backbone rank lift for such observations. NTD-PL first forms a shared Tucker latent tensor and then maps each latent entry through a finite polynomial link. The model contains standard Tucker as the degree-one case and adds only a small number of scalar link coefficients. At the same time, powers of the latent tensor induce tied high-order interactions among the same Tucker factors, giving a compact rank-lift without introducing a full high-order tensor parameter.

We study the representation power of this model through approximation and separation results, and derive a masked learning objective for reconstruction and completion. The resulting solver alternates between Tucker backbone updates and a small ridge-regression step for the link coefficients, with degree continuation for stability. Experiments on controlled tensors, CAVE hyperspectral completion and reconstruction, and external HSI scenes show that NTD-PL improves matched-rank Tucker when the residual has a coherent nonlinear component. Rank-lift and Link Refresh diagnostics show that the gain is tied to jointly reshaping the Tucker backbone under a shared nonlinear link, rather than to a small ordinary rank increase alone.

1 Introduction

Tensor decompositions provide compact models for multiway data in scientific sensing, imaging, and signal processing [Kolda and Bader, 2009, Sidiropoulos et al., 2017, Liu et al., 2013]. Tucker decomposition is a standard choice because it separates a small core from mode-wise factors, but its observation model is linear in the reconstructed latent entry. This can be too restrictive when a low-rank latent signal passes through saturation, response curves, calibration effects, or material-dependent nonlinearities before it is measured [Debevec and Malik, 1997, Grossberg and Nayar, 2004, Bioucas-Dias et al., 2012, Heylen et al., 2014]. After such an entry-level response, a simple latent Tucker tensor may appear higher-rank in the observed domain.

A direct remedy is to increase Tucker rank, but this expands the core and factors throughout the model. More flexible nonlinear tensor decoders based on kernels, Gaussian processes, or neural networks can also help, but they often make the relation to the Tucker backbone less transparent [Xu et al., 2011, Zhe et al., 2016, Liu et al., 2017, Saragadam et al., 2024]. We ask a narrower question: can a small observation link, learned jointly with the Tucker factors, make a tight multilinear rank budget more effective when the dominant residual is shared and entrywise?

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We instantiate this idea with a polynomial link, a classical low-dimensional tool for modeling nonlinear responses [Volterra, 1959, Boyd and Chua, 1985]. Let

$$\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)}]\!]$$

be a Tucker latent tensor. Instead of fitting the observation directly by $\mathcal{S}$, we model

$$\widehat{\mathcal{Y}} = f_{\boldsymbol{\beta}}(\mathcal{S}) = \sum_{p=0}^{P} \beta_p \mathcal{S}^{\odot p},$$

where powers are entrywise. We call the model Nonlinear Tucker Decomposition with Polynomial Links (NTD-PL). It contains Tucker as the degree-one case and adds only $P+1$ scalar coefficients, while the powers $\mathcal{S}^{\odot p}$ induce tied high-order interactions among the same core and factors. Thus the link is not an independent high-rank correction; it is a compact way to expose nonlinear structure already organized by the Tucker backbone.

This viewpoint is central to the paper. NTD-PL is best understood as a Tucker enhancement and shared-backbone rank lift, not as a generic nonlinear decoder. A degree-p power of the latent tensor behaves like an ordinary Tucker tensor whose factors are polynomial feature lifts of the original factors, but those lifted features remain tied to one learned backbone. The new degrees therefore reshape the low-rank geometry through a shared observation link, rather than adding an untied high-rank tensor block.

Our claims are intentionally scoped. NTD-PL is not meant to replace rank selection or general nonlinear tensor decoders. It targets settings where a shared entry-level response explains coherent residual structure left by a low-rank Tucker model.

Contributions. We make the following contributions.

- We propose NTD-PL, a Tucker model with a jointly learned polynomial observation link. Standard Tucker is recovered as the degree-one case, while higher degrees act as a Tucker enhancement and shared-backbone rank lift of the same backbone.
- We characterize the model class through Veronese rank-expansion, separation from standard Tucker at the same backbone rank, and approximation results for shared nonlinear observation maps.
- We derive a masked learning objective for reconstruction and completion, and use an alternating solver with a closed-form ridge update for the link coefficients.
- We evaluate the mechanism on controlled tensors and hyperspectral data, including completion, reconstruction, rank-lift, and Link Refresh residual-yield diagnostics, including predicted boundary cases.

2 Related Work

Low-rank tensor models. CP, Tucker, tensor train, and tensor ring decompositions provide compact representations for multiway data [Kolda and Bader, 2009, De Lathauwer et al., 2000, Oseledets, 2011, Zhao et al., 2016]. Tucker is the closest baseline for NTD-PL because its core and mode factors give the same multilinear backbone. Rather than changing ranks, factor structures, or regularization, we keep this backbone fixed and change the observation map from the latent tensor to the measured entries.

Nonlinear tensor factorization. Nonlinear tensor methods use generalized likelihoods, kernels, Gaussian processes, neural decoders, or nonlinear tensor networks [Hong et al., 2020, Xu et al., 2011, Zhe et al., 2016, Pan et al., 2020, Saragadam et al., 2024, Varolgunes et al., 2023]. These models can be more expressive than a scalar polynomial link, but they often trade away the simple finite-dimensional relation to Tucker. NTD-PL is complementary: it keeps the nonlinearity shared and entrywise so that its effect can be analyzed as a tied rank lift of the same Tucker factors.

Polynomial feature lifts and high-order interactions. Polynomial and Volterra-style expansions are classical tools for representing nonlinear responses [Volterra, 1959, Boyd and Chua, 1985]. Related matrix and tensor methods use polynomial feature maps or symmetric high-order tensors to increase expressivity [Zhai et al., 2024, Liu et al., 2015, Deng et al., 2016]. Such lifts can expose nonlinear structure, but explicit high-order parameters often increase dimension and contraction cost. NTD-PL can be viewed as a shared-backbone compression of these interactions: the lifted features are induced by powers of one learned Tucker latent tensor, rather than by an independent high-order tensor parameter.

3 Model

3.1 Model and Observation Assumption

Let $\mathcal{Y} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ be an observed tensor and let

$$\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(N)}]\!], \qquad \mathcal{X} \in \mathbb{R}^{R_1 \times \cdots \times R_N}, \quad \mathbf{A}^{(n)} \in \mathbb{R}^{I_n \times R_n}. \tag{1}$$

NTD-PL treats $\mathcal{S}$ as a shared low-rank latent tensor and models each observed entry through a scalar polynomial response:

$$Y_{\mathbf{i}} = f_{\boldsymbol{\beta}}(S_{\mathbf{i}}) + \varepsilon_{\mathbf{i}}, \qquad f_{\boldsymbol{\beta}}(s) = \sum_{p=0}^{P} \beta_p s^p, \qquad \mathbf{i} = (i_1, \ldots, i_N). \tag{2}$$

Equivalently, the tensor prediction is

$$\widehat{\mathcal{Y}} = f_{\boldsymbol{\beta}}(\mathcal{S}) = \sum_{p=0}^{P} \beta_p \mathcal{S}^{\odot p}, \qquad \boldsymbol{\beta} \in \mathbb{R}^{P+1}. \tag{3}$$

The constant term β_0 captures affine response shifts, the linear term $\beta_1 \mathcal{S}$ is the usual Tucker channel, and terms with $p \geq 2$ activate nonlinear channels on the same latent tensor. Standard Tucker is recovered by setting $\beta_1 = 1$ and $\beta_p = 0$ for $p \neq 1$. Thus NTD-PL changes the entry-generation map while leaving the Tucker backbone and its multilinear ranks fixed.

3.2 Shared-Backbone High-Order Interactions

Although $f_{\boldsymbol{\beta}}$ is scalar, the term $\mathcal{S}^{\odot p}$ is not merely an independent output calibration. We arrive at it by first defining an explicit p-fold Tucker extension: instead of mapping one latent index per mode, the mode-n projection is a higher-order tensor $\mathcal{A}^{(n,p)} \in \mathbb{R}^{I_n \times R_n^p}$ that couples p latent indices. The case $p = 1$ is ordinary Tucker, while $p = 2$ gives a quadratic Tucker-style interaction between two latent Tucker tuples. Appendix A.1 gives the formal entrywise construction and tensor-network view.

The explicit construction is a reference model, not the parameterization we train. Its projection tensors grow as $I_n R_n^p$, and each degree would add a new set of mode projections. NTD-PL keeps the same high-order interaction pattern by tying the p-fold projection to the original Tucker factor:

$$\mathcal{A}^{(n,p)}_{i_n, r_n^{(1)}, \ldots, r_n^{(p)}} = \prod_{q=1}^{p} A^{(n)}_{i_n, r_n^{(q)}}.$$

Under this tying, the degree-p channel is exactly $S_{\mathbf{i}}^p$. Thus it still couples p latent Tucker index tuples, but all copies share the same mode factors. Across degrees, the only new free parameters are the scalar coefficients $\{\beta_p\}$, which mix these tied channels.

The latent tensor $\mathcal{S}$ and the coefficients $\boldsymbol{\beta}$ are fitted jointly, so the low-rank backbone is optimized together with the polynomial channels. Section 4 analyzes the induced rank expansion and the resulting separation from standard Tucker.

3.3 Masked Least-Squares Objective

Let $\Omega \in \{0,1\}^{I_1 \times \cdots \times I_N}$ denote the observed-entry mask and let $|\Omega| = \sum_{\mathbf{i}} \Omega_{\mathbf{i}}$. We estimate the core, factors, and link coefficients by the regularized masked objective

$$\min_{\mathcal{X}, \{\mathbf{A}^{(n)}\}, \boldsymbol{\beta}} \frac{1}{2|\Omega|} \left\| \Omega \odot (\mathcal{Y} - f_{\boldsymbol{\beta}}(\mathcal{S})) \right\|_F^2 + \frac{\lambda_X}{2} \left\| \mathcal{X} \right\|_F^2 + \sum_{n=1}^{N} \frac{\lambda_n}{2} \left\| \mathbf{A}^{(n)} \right\|_F^2 + \frac{\lambda_\beta}{2} \left\| \boldsymbol{\beta} \right\|_2^2. \tag{4}$$

The same formulation covers full reconstruction by taking $\Omega \equiv \mathbf{1}$ and tensor completion by restricting the loss to the observed entries. The quadratic penalties regularize the Tucker backbone and the low-dimensional polynomial link.

4 Theory

The theory addresses the rank-lift view raised by the model definition. A scalar link has few free parameters, but its powers can induce ordinary Tucker tensors with larger multilinear ranks. We first make this induced structure explicit, then give a separation from matched-rank Tucker and an approximation route for shared nonlinear observation maps.

4.1 Rank Expansion and Separation

Proposition 1 (Shared-backbone rank expansion)**.** *Let $\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(N)}]\!]$ have Tucker rank at most $(R_1, \ldots, R_N)$. For every integer $p \geq 1$, $\mathcal{S}^{\odot p}$ admits a Tucker representation with mode ranks at most*

$$\left(\binom{R_1 + p - 1}{p}, \ldots, \binom{R_N + p - 1}{p}\right). \tag{5}$$

More explicitly, the ordinary Tucker factors of $\mathcal{S}^{\odot p}$ can be chosen as Veronese lifts of the original factors. For $\mathbf{A} \in \mathbb{R}^{I \times R}$, define $\mathbf{A}^{\langle p \rangle} \in \mathbb{R}^{I \times D}$, $D = \binom{R+p-1}{p}$, by

$$A^{\langle p \rangle}_{i,\alpha} = \prod_{r=1}^{R} A^{\alpha_r}_{i,r}, \tag{6}$$

where $\alpha \in \mathbb{N}^R$ ranges over $|\alpha| = p$. Then $\mathcal{S}^{\odot p}$ factors through $\mathbf{A}^{(1)\langle p \rangle}, \ldots, \mathbf{A}^{(N)\langle p \rangle}$.

This proposition makes the tying in Section 3.2 explicit in standard Tucker language: a degree-p channel is an ordinary Tucker tensor whose mode factors are polynomial feature lifts of the same backbone factors. The link therefore changes rank structure through tied features, not through an independent high-order parameter block.

Theorem 1 (Tensorial separation from standard Tucker)**.** *Let $N \geq 3$, $R \geq 2$, $p \geq 2$, and suppose $I_n \geq D$ for every n, where $D = \binom{R+p-1}{p}$. Then there exists an order-N tensor $\mathcal{Y} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ that is representable by a degree-p NTD-PL model with Tucker backbone rank $(R, \ldots, R)$, but whose exact standard Tucker rank is*

$$(D, \ldots, D). \tag{7}$$

Equivalently, a degree-p polynomial link can generate tensors whose multilinear ranks are $(\binom{R+p-1}{p}, \ldots, \binom{R+p-1}{p})$ from a shared Tucker backbone of rank $(R, \ldots, R)$.

The proof in Appendix B uses a CP-diagonal Tucker backbone and generic full-rank Veronese feature matrices. The theorem is an existence separation, not a universality claim over all high-rank Tucker tensors: it identifies a structured high-rank subset generated by a low-rank shared backbone.

4.2 Approximation Under Nonlinear Observations

Theorem 2 (Polynomial generation)**.** *Suppose $\mathcal{Y} = g(\mathcal{S}^\star)$, where $\mathcal{S}^\star$ has Tucker rank at most $(R_1, \ldots, R_N)$ and g is a polynomial of degree at most P. Then degree-P NTD-PL with the same Tucker rank can represent $\mathcal{Y}$ exactly.*

Theorem 3 (Analytic link approximation)**.** *Suppose $\mathcal{Y} = g(\mathcal{S}^\star)$, where $\mathcal{S}^\star$ has Tucker rank at most $(R_1, \ldots, R_N)$ and $\|\mathcal{S}^\star\|_\infty \leq B$. If g is analytic on the complex disk $D(0, \rho)$ with $B < \rho$, and $M_\rho = \sup_{|z|=\rho} |g(z)|$, then there exists a degree-P NTD-PL model with the same Tucker backbone satisfying*

$$\frac{1}{\sqrt{M}} \left\|\mathcal{Y} - \widehat{\mathcal{Y}}\right\|_F \leq M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}, \qquad M = \prod_{n=1}^{N} I_n. \tag{8}$$

Thus the same finite link that induces a tied rank lift also gives a direct approximation route for smooth entrywise responses on a bounded latent range. The statement uses the simple Taylor truncation bound; sharper polynomial approximation bounds can replace it under stronger assumptions.

Together, the results clarify the scope of NTD-PL. Polynomial links can approximate shared nonlinear observation maps while keeping the same latent Tucker rank, and their powers induce Veronese-lifted Tucker factors. This explains how a scalar link can change multilinear rank structure without introducing untied high-order tensor parameters.

5 Optimization

We optimize the masked objective in Section 3.3 by alternating between the scalar link and the Tucker backbone. The nonlinear part is low-dimensional once the latent Tucker tensor is fixed, while the backbone update remains a standard differentiable Tucker fitting step through the current link. We call the two stages *Link Refresh* and *Linked-Backbone Descent*.

Link Refresh. Fix the current latent tensor

$$S = [\![\mathcal{X}; \mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(N)}]\!]. \tag{9}$$

For active degree $Q \le P$, updating the polynomial coefficients reduces to a small masked ridge regression,

$$\min_{\boldsymbol{\beta}_{0:Q}} \frac{1}{2|\Omega|} \left\| y_\Omega - \boldsymbol{\Phi}_{\Omega,Q}(S)\boldsymbol{\beta}_{0:Q} \right\|_2^2 + \frac{\lambda_\beta}{2} \left\| \boldsymbol{\beta}_{0:Q} \right\|_2^2, \tag{10}$$

where $\boldsymbol{\Phi}_{\Omega,Q}(S)$ contains the observed powers $[1, s, \ldots, s^Q]$. This step is cheap because it solves only a $(Q+1) \times (Q+1)$ linear system. It also gives an interpretable view of the link: once the backbone exposes a one-dimensional latent value at each entry, the refresh estimates the best shared scalar response on the observed entries.

Linked-Backbone Descent. With $\boldsymbol{\beta}$ fixed, the Tucker parameters are updated through the current nonlinear observation map. The masked residual is first propagated to the latent tensor as

$$T = \frac{1}{|\Omega|} \Omega \odot (f_{\boldsymbol{\beta}}(S) - Y) \odot f'_{\boldsymbol{\beta}}(S), \tag{11}$$

and then backpropagated through the Tucker reconstruction to update $\mathcal{X}$ and $\{\mathbf{A}^{(n)}\}$. In implementation we use Adam for these first-order steps, but the update itself is just standard Tucker reverse contractions composed with the current link. The backbone is therefore not calibrated after fitting; it is jointly reshaped under the shared nonlinear link.

Stabilization. Directly activating all polynomial degrees from the start is brittle, because higher powers of S amplify scale ambiguity. We therefore begin from the Tucker-compatible degree-one model and activate higher degrees gradually. When a new degree is switched on, its coefficient is initialized at zero, so the model expands from the current lower-degree solution without changing the prediction abruptly. We also normalize Tucker factor scales and absorb the corresponding scales into the core, which fixes a Tucker gauge without changing S. For the coefficient solve, the implementation may regress on a centered and scaled latent variable and then map the solution back to the original power basis.

Cost. The dominant cost is the same Tucker forward and reverse contractions used by standard Tucker optimization at the chosen rank. The polynomial link adds only $O(P|\Omega|)$ elementwise work for evaluating $f_{\boldsymbol{\beta}}(S)$ and $f'_{\boldsymbol{\beta}}(S)$, plus a small $(P+1) \times (P+1)$ ridge solve. Since the experiments use small P, the overhead is modest relative to the Tucker contractions. Full gradients, stabilization details, and pseudocode are given in Appendix A.2.

Masked-entry uniform convergence. The masked objective also admits a standard sampling interpretation. Let $m = |\Omega|$ and let $\Theta \subset \mathbb{R}^d$ be a bounded NTD-PL parameter set, where $d = \prod_{n=1}^N R_n + \sum_{n=1}^N I_n R_n + (P+1)$. If the entrywise losses are bounded by $0 \le \ell_{\mathbf{i}}(\theta) \le B_\ell$ and G-Lipschitz in θ, then with probability at least $1-\delta$,

$$\sup_{\theta\in\Theta} \left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \le 2G\varepsilon + B_\ell \sqrt{\frac{d\log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}, \tag{12}$$

for every $0 < \varepsilon \le B_\Theta$. This is a standard uniform convergence bound, not a sharp minimax result, but it clarifies that masked training controls the full-entry risk with a complexity term determined by the tied NTD-PL parameter count. The proof is given in Appendix B.4.

6 Experiments

This section asks three questions: (i) Does NTD-PL behave as expected under controlled nonlinear residuals? (ii) Does the polynomial link enhance a Tucker backbone in reconstruction and held-out completion? (iii) Can we diagnose when the enhancement appears across datasets? Unless stated otherwise, real tensors are globally max-normalized and used without spatial downsampling or cropping. RMSE and SAM are used for reconstruction; RMSE* and SAM* are computed only on held-out missing entries for completion. NTD-PL is compared mainly against Tucker at the same multilinear rank, except in the explicit rank-lift ablation. We also use a Link Refresh diagnostic score, D_{link}: after fitting Tucker, run the cheap scalar Link Refresh ridge solve once on the Tucker latent tensor, and measure the residual reduction on task-specific evaluation entries. Larger D_{link} means the Tucker residual is more compatible with a shared nonlinear link.

6.1 Controlled Nonlinear Residuals

This section isolates the mechanism on synthetic tensors. We generate observations as

$$Y_\alpha = \sqrt{1-\alpha}S^\star + \sqrt{\alpha}\widetilde{R}^\star,$$

where $S^\star$ is a low-rank Tucker tensor and $\widetilde{R}^\star$ is the normalized component of a scalar nonlinear response orthogonal to $S^\star$. Thus α controls nonlinear residual energy rather than the raw amplitude of a scalar nonlinearity. When the source is linear or nearly linear, NTD-PL should collapse toward Tucker-like behavior; as nonlinear residual energy increases, it should separate from linear tensor baselines. The alpha sweep in Figure 1 shows this transition, while the degree-gain heatmap in Figure 2 ties the gain to the expressive order of the polynomial link. Degree continuation and coefficient-contribution details are deferred to Appendix C.1, which also reports the linear consistency negative control.

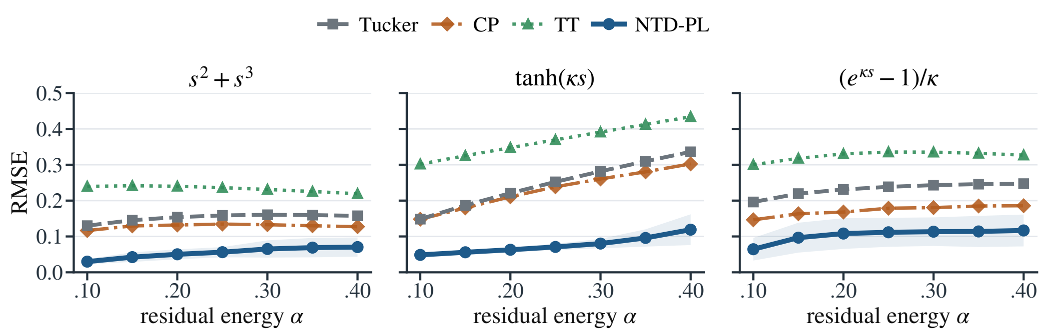

Figure 1: Controlled nonlinear alpha sweep. Each panel uses a different scalar response to define the orthogonal residual; α sets nonlinear residual energy, and lower RMSE is better.

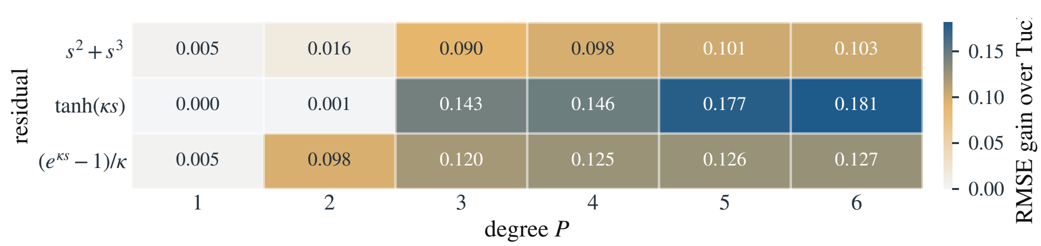

Figure 2: Controlled nonlinear degree-gain heatmap at $\alpha = 0.25$. Cell values are Tucker minus NTD-PL RMSE, so larger values favor NTD-PL.

6.2 Tucker Enhancement on Real Hyperspectral Tensors

This section asks whether NTD-PL enhances a Tucker backbone on real hyperspectral tensors, both when all entries are observed and when evaluation is performed on held-out missing entries.

Representation enhancement. Full-observation reconstruction is a representation-capacity check, not a generalization test. Figure 3 shows that NTD-PL improves both RMSE and SAM on CAVE at matched Tucker backbone rank and similar parameter budget. The advantage shrinks as Tucker rank increases, which is expected: ordinary multilinear capacity already explains more of the tensor. The rank-lifted Tucker ablation in Table 1 clarifies the mechanism further. Tucker can buy back part of the RMSE gap by spending more rank, so the gain is not explained by a small ordinary rank increase alone. They support the view that NTD-PL improves a tight Tucker backbone through a jointly fitted shared link. The joint link-family smoke test in Appendix C.4 further suggests that the effect is tied to joint scalar-link fitting rather than to the polynomial basis alone.

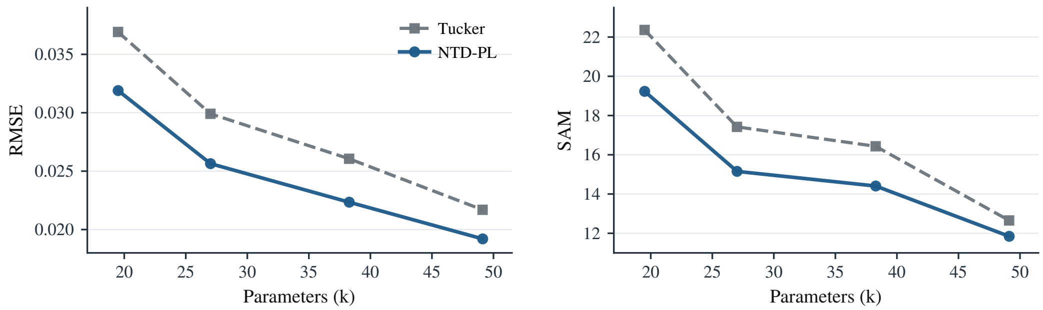

Figure 3: CAVE full-observation reconstruction as a representation-capacity check. Each point averages 15 scenes at a matched Tucker backbone rank. The plotted backbone ranks are $(18, 18, 3)$, $(24, 24, 4)$, $(33, 33, 4)$, and $(40, 40, 5)$. Unlike completion, this measures representational error when all entries are observed.

Table 1: Rank-lifted Tucker comparison on CAVE reconstruction. For each NTD-PL backbone rank, we increase Tucker's spatial ranks while keeping the spectral rank fixed, and report the first Tucker rank that matches the NTD-PL RMSE.

NTD-PL reference				First Tucker matching RMSE				Extra params
Rank	Params	RMSE↓	SAM↓	Rank	Params	RMSE↓	SAM↓	
$(18, 18, 3)$	20k	0.0319	**19.23**	$(28, 28, 3)$	31k	**0.0316**	21.41	$+59.5\%$
$(24, 24, 4)$	27k	0.0256	**15.15**	$(35, 35, 4)$	41k	**0.0254**	16.26	$+51.3\%$
$(33, 33, 4)$	38k	0.0223	**14.41**	$(49, 49, 4)$	60k	**0.0222**	15.28	$+56.5\%$

Held-out enhancement. Masked completion is the stronger test because it asks whether the learned low-rank geometry transfers to unobserved entries. Table 2 reports CAVE completion at several random mask rates, with RMSE* and SAM* computed only on missing entries. Under the same rank budget as Tucker, NTD-PL gives consistent held-out RMSE* gains and usually improves SAM* as well. The paired sign and Wilcoxon tests in Appendix C.2 show the same pattern scene by scene.

Table 2: Full-resolution CAVE completion at rank $(33, 33, 4)$ under random masks. Here ρ denotes the missing-entry rate. RMSE* and SAM* are computed on missing entries. Tucker and NTD-PL columns report mean $\pm$ standard deviation over scenes; gains are paired relative improvements over Tucker.

ρ	RMSE*↓			SAM*↓		
	Tucker	NTD-PL	Gain	Tucker	NTD-PL	Gain
0.1	0.0263±0.0179	**0.0229±0.0158**	12.9%	11.82±5.33	**8.36±3.38**	29.3%
0.3	0.0263±0.0179	**0.0230±0.0158**	12.5%	14.75±6.33	**12.47±5.06**	15.5%
0.5	0.0264±0.0179	**0.0230±0.0158**	12.9%	15.56±6.85	**13.48±5.75**	13.4%
0.7	0.0266±0.0180	**0.0219±0.0128**	17.7%	16.02±7.21	**13.79±5.43**	13.9%

6.3 Diagnosing Nonlinear Low-Rank Structure Across Data

NTD-PL is not expected to help uniformly on all tensors. Its gain should appear when a matched-rank Tucker fit leaves a coherent residual that a shared scalar link can reduce. We denote this diagnostic by

D_{link}: after fitting Tucker, we freeze the Tucker backbone, run one scalar Link Refresh ridge solve, and measure the residual-energy reduction on the relevant evaluation entries. Larger D_{link} indicates a stronger shared entry-level nonlinear residual:

$$D_{\text{link}} = 10 \log_{10} \frac{\|Y_{\text{eval}} - \widehat{Y}_T\|_F^2}{\|Y_{\text{eval}} - \widehat{Y}_{\text{link}}\|_F^2},$$

where $\widehat{Y}_T$ is the Tucker output and $\widehat{Y}_{\text{link}}$ is the output after the one-step refresh.

For HSI, we use natural scene-level tensor units rather than artificial patches. CAVE already provides multiple scene tensors, so Figure 4 evaluates the scene-level relationship between D_{link} and NTD-PL held-out RMSE* gain.

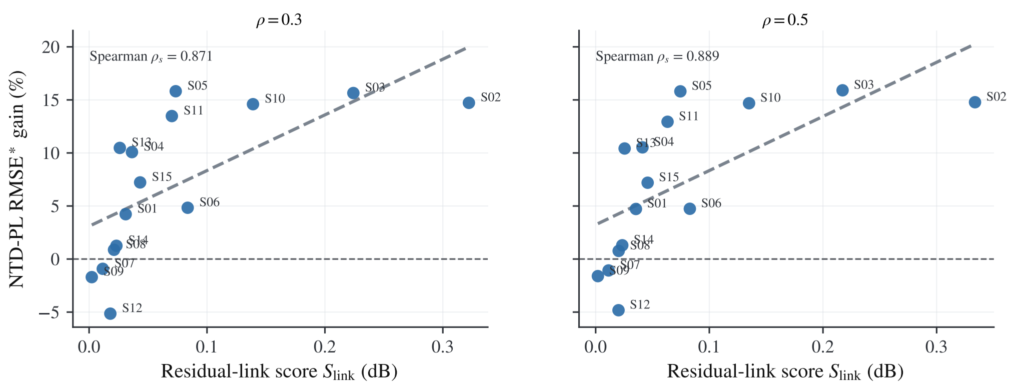

Figure 4: CAVE completion at $\rho \in \{0.3, 0.5\}$: scene-level NTD-PL RMSE* gain versus Link Refresh diagnostic D_{link}. The left and right panels use $\rho = 0.3$ and $\rho = 0.5$, respectively, and each panel annotates its Spearman coefficient.

Figure 4 first validates the diagnostic on CAVE completion: scenes with higher D_{link} receive larger held-out RMSE* gains. We then extend the same protocol beyond HSI. For natural images, each image is treated as one tensor unit; for object-view datasets, each object or instance forms a multi-view tensor unit. We do not concatenate unrelated images or objects into a giant tensor, since D_{link} tests whether a single shared scalar link is coherent within one unit. External HSI scenes are also checked as whole cubes; Jasper Ridge, Samson, and Urban have positive Link Refresh scores, while Cuprite is a near-zero D_{link} boundary case.

Table 3: Cross-domain Link Refresh diagnosis. Each unit is independently fitted by Tucker and NTD-PL at matched backbone rank. D_{link} is computed by one frozen-backbone Link Refresh step and diagnostic tiers are assigned from D_{link}, not from NTD-PL gain. Natural images use per-image units; object-view datasets use per-object or per-instance view tensors.

Domain	Dataset	Unit	#Units	Med. D_{link}	Spearman	Med. gain	Low/High gain
HSI	CAVE completion	scene	15	–	0.889	–	$-1.09/13.18\%$
HSI	External HSI	whole cube	4	–	–	–	–
Natural images	CBSD68	image	16	0.075	0.835	5.77%	$3.87/12.44\%$
Natural images	CIFAR-10	image	1000	0.008	0.934	2.35%	$-0.01/7.85\%$
Object-view	COIL-100	object-view tensor	100	0.208	0.932	5.40%	$2.77/9.14\%$
Object-view	smallNORB	instance-view tensor	25	0.264	0.945	9.84%	$6.92/14.16\%$

Table 3 summarizes the cross-domain diagnosis. Natural images show weaker shared-link yield, especially CIFAR-10, whose median D_{link} is close to zero and whose low-yield tercile has almost no gain. Object-view tensors show clearer effective units: COIL-100 and smallNORB have larger median D_{link}, stronger rank-matched gains, and high-yield terciles with larger improvements. These results support the scoped claim that NTD-PL helps when the Tucker residual contains shared link-like structure, rather than uniformly across all visual tensors. Representative units selected only by D_{link} are reported in Appendix C.7: high-D_{link} object-view units such as COIL object 91 and smallNORB category 2 instance 7 show large gains, while low-D_{link} boundary units show small or negative gains.

7 Limitations

The diagnostic experiments in Section 6 also expose the main limitation. NTD-PL assumes that the residual left by a low-rank Tucker backbone is well described by a shared entry-level nonlinear response. It is therefore not a universal tensor model: when Tucker already fits well at the chosen rank, or when residual nonlinearities vary strongly across space, spectra, or modes, a single scalar link may add little.

The method also does not remove rank selection. Increasing Tucker rank can recover part of the RMSE gap, and NTD-PL inherits Tucker's nonconvexity and initialization sensitivity. Although the link adds few parameters, evaluating it over all observed entries still increases runtime relative to plain Tucker.

8 Conclusion

We introduced NTD-PL, a Tucker enhancement with a shared polynomial observation link. Rather than replacing multilinear low-rank structure, NTD-PL acts as a shared-backbone rank lift: powers of one learned latent tensor express structured residuals left by a linear Tucker fit without adding an independent high-rank correction.

The experiments support the same scoped interpretation. Controlled tensors, held-out CAVE completion, reconstruction mechanism checks, Link Refresh diagnostics, and external HSI scenes show that the link helps when a shared nonlinear residual is plausible under a tight rank budget, while also revealing predicted near-linear boundary cases. Adaptive links and broader tensor modalities are natural next steps.

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A Additional Model Details

A.1 Explicit p-Fold Tucker Motivation

This section expands the motivation behind the explicit p-fold Tucker channel in Section 3.2. The construction is best viewed as the tensor analogue of adding polynomial interactions to a linear map. In the matrix case, one may move from a linear model $\mathbf{y} = \mathbf{A}\mathbf{x}$ to a quadratic model

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathcal{A}(\mathbf{x}, \mathbf{x}),$$

where the second term couples pairs of latent coordinates. The corresponding question for Tucker is: if standard Tucker is the multilinear low-rank map for multiway data, what is the direct p-th order interaction analogue?

For a Tucker backbone, an explicit p-fold interaction can be written as

$$\widetilde{Y}_{\mathbf{i}}^{(p)} = \sum_{\mathbf{r}^{(1)},\dots,\mathbf{r}^{(p)}} \left(\prod_{q=1}^{p} \mathcal{X}_{r_1^{(q)},\dots,r_N^{(q)}} \right) \prod_{n=1}^{N} \mathcal{A}^{(n,p)}_{i_n, r_n^{(1)},\dots,r_n^{(p)}}. \tag{13}$$

Here each mode projection depends on p latent indices instead of one. For $p = 1$, this reduces to the ordinary Tucker entry formula. For $p = 2$, the same core appears twice and the mode projections couple two latent tuples, giving a quadratic Tucker-style channel. Figure 5 illustrates this progression.

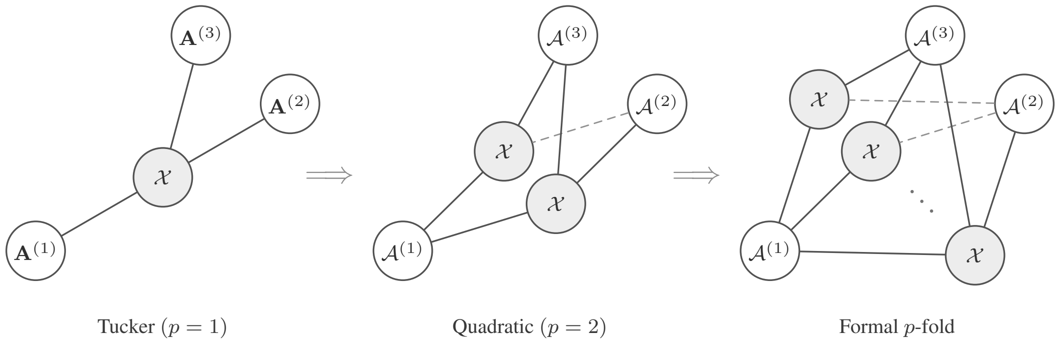

Figure 5: Tensor-network view of a formal p-fold Tucker extension. The construction keeps the Tucker idea of mode-wise generation, but lets each mode projection couple multiple latent index tuples.

The explicit construction is useful as a reference model, but it is not the parameterization used by NTD-PL. Its mode projections $\mathcal{A}^{(n,p)}$ have size $I_n R_n^p$, so a direct implementation introduces new high-order projection tensors and rapidly increases storage and contraction cost. NTD-PL keeps the same high-order interaction pattern by tying the projection tensor to the original Tucker factor:

$$\mathcal{A}^{(n,p)}_{i_n, r_n^{(1)}, \ldots, r_n^{(p)}} = \prod_{q=1}^{p} A^{(n)}_{i_n, r_n^{(q)}}. \tag{14}$$

Substituting (14) into (13) gives exactly $\mathcal{S}^{\odot p}$, where $\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(N)}]\!]$. Thus the degree-$p$ NTD-PL channel is a shared-projection version of the explicit p-fold Tucker prototype.

This is the modeling role of the p-fold Tucker prototype: it defines the high-order Tucker interaction that NTD-PL compresses. The final model does not train the untied tensors $\mathcal{A}^{(n,p)}$; it keeps a single Tucker backbone and adds only the polynomial coefficients $\{\beta_p\}$.

A.2 Optimization Details

This section gives the update equations behind the solver summarized in Section 5. The implementation follows the same structure: one Tucker reconstruction per iteration, a fused evaluation of the polynomial and its derivative, first-order updates of the backbone, and periodic ridge-regression updates of the link coefficients.

Backbone gradients. For a fixed $\boldsymbol{\beta}$, define

$$f'_{\boldsymbol{\beta}}(\mathcal{S}) = \sum_{p=1}^{P} p \beta_p \mathcal{S}^{\odot (p-1)}. \tag{15}$$

The gradient of the normalized masked data term with respect to the latent tensor $\mathcal{S}$ is

$$\mathbf{T} = \frac{1}{|\Omega|} \Omega \odot (f_{\boldsymbol{\beta}}(\mathcal{S}) - \mathcal{Y}) \odot f'_{\boldsymbol{\beta}}(\mathcal{S}). \tag{16}$$

Backpropagating $\mathbf{T}$ through the Tucker reconstruction gives the core gradient

$$\nabla_{\mathcal{X}} \mathcal{L} = \mathbf{T} \times_1 \mathbf{A}^{(1)\top} \times_2 \cdots \times_N \mathbf{A}^{(N)\top} + \lambda_X \mathcal{X}. \tag{17}$$

For the n-th factor, let

$$\mathbf{Z}^{(n)} = \left[\mathcal{X} \times_1 \mathbf{A}^{(1)} \cdots \times_{n-1} \mathbf{A}^{(n-1)} \times_{n+1} \mathbf{A}^{(n+1)} \cdots \times_N \mathbf{A}^{(N)} \right]_{(n)}. \tag{18}$$

Then

$$\nabla_{\mathbf{A}^{(n)}} \mathcal{L} = \mathbf{T}_{(n)} \mathbf{Z}^{(n)\top} + \lambda_n \mathbf{A}^{(n)}. \tag{19}$$

The experiments use Adam for these first-order steps. This choice is not part of the model definition; any comparable first-order optimizer can use the same gradients.

Link update. With $\mathcal{S}$ fixed and active degree $Q = P_{\text{act}}$, the coefficient subproblem is the ridge regression in (10). Its closed-form solution is

$$\boldsymbol{\beta}^{\star}_{0:Q} = \left(\boldsymbol{\Phi}_{\Omega,Q}^{\top}\boldsymbol{\Phi}_{\Omega,Q} + |\Omega|\lambda_{\beta}\mathbf{I}\right)^{-1}\boldsymbol{\Phi}_{\Omega,Q}^{\top}\mathbf{y}_{\Omega}. \tag{20}$$

Equivalently, one may absorb $|\Omega|$ into the ridge parameter and solve the usual unnormalized normal equations. If the constant term is disabled, the first column is removed and β_0 is fixed at zero. In the code, this small system is solved either from an explicit Vandermonde design or from moment equations built from cached powers.

Stabilization and continuation. High powers of $\mathcal{S}$ can be sensitive to scale. The implementation therefore normalizes factor columns after backbone updates and absorbs the corresponding scales into the core; this is a gauge fixing of the Tucker parameterization and does not change $\mathcal{S}$. For the coefficient solve, it may regress on the centered and scaled latent variable $(S-\mu)/\sigma$, then convert the solution back to the original power basis of S.

The active degree follows the schedule

$$t_k = \operatorname{round}\left(\frac{kT}{P}\right), \qquad k = 1, \dots, P-1.$$

When degree $p+1$ is activated, the previous coefficients and Tucker factors are kept and the new coefficient β_{p+1} is initialized at zero.

Cost. The dominant cost is forming the Tucker reconstruction and its reverse contractions on the observed tensor. For dense tensors this is comparable to standard Tucker optimization at the same rank, plus $O(P|\Omega|)$ work to evaluate the scalar polynomial and its derivative. The link update forms a Vandermonde design or the corresponding moment equations and solves a $(P+1)\times(P+1)$ linear system, with direct cost $O(P^2|\Omega| + P^3)$. Since P is small, this cost is negligible compared with the Tucker contractions.

Algorithm 1 gives the full procedural summary.

Algorithm 1 Masked NTD-PL optimization

```
Require: observed tensor 𝒴, mask Ω, rank (R_1, ..., R_N), degree P
initialize 𝒳, {A^(n)} from Tucker on the masked tensor
initialize active degree P_act ← 1 and β = (0, 1)
for t = 1, ..., T do
    form 𝒮 = ⟦𝒳; A^(1), ..., A^(N)⟧
    if a link update is scheduled then
        update active coefficients β_{0:P_act} by ridge regression on {(S_i, Y_i) : Ω_i = 1}
    end if
    compute f_β(𝒮) and f'_β(𝒮)
    take first-order steps on 𝒳 and {A^(n)} using the masked gradient
    normalize factor scales and absorb them into the core
    increase P_act according to the continuation schedule, initializing the new coefficient at zero
end for
return 𝒳, {A^(n)}, β
```

B Additional Proofs

B.1 Polynomial and Analytic Approximation

Proof of Theorem 2. Since g has degree at most P, write

$$g(s) = \sum_{p=0}^{P} c_p s^p. \tag{21}$$

Take the NTD-PL latent tensor to be the true latent tensor, $\mathcal{S} = \mathcal{S}^\star$, and set $\beta_p = c_p$. Then, entrywise,

$$\widehat{\mathcal{Y}}_{i_1,\dots,i_N} = \sum_{p=0}^{P} \beta_p \left(\mathcal{S}^\star_{i_1,\dots,i_N}\right)^p = g\left(\mathcal{S}^\star_{i_1,\dots,i_N}\right) = \mathcal{Y}_{i_1,\dots,i_N}. \tag{22}$$

Thus $\widehat{\mathcal{Y}} = \mathcal{Y}$. $\square$

Lemma 1 (Scalar-to-tensor approximation)**.** *Let $\|\mathcal{S}^\star\|_\infty \le B$. For any scalar function g and any degree-P polynomial q_P,*

$$\|g(\mathcal{S}^\star) - q_P(\mathcal{S}^\star)\|_F \le \sqrt{\prod_{n=1}^{N} I_n} \sup_{|s|\le B} |g(s) - q_P(s)|. \tag{23}$$

Proof. Every entry of $\mathcal{S}^\star$ lies in $[-B, B]$, so every entrywise approximation error is bounded by $\sup_{|s|\le B} |g(s) - q_P(s)|$. Squaring, summing over all entries, and taking the square root gives the claim. $\square$

Proof of Theorem 3. Let $g(s) = \sum_{k=0}^{\infty} a_k s^k$ be the Taylor expansion of g around zero, and choose $q_P(s) = \sum_{k=0}^{P} a_k s^k$. By Cauchy's estimate,

$$|a_k| \le \frac{M_\rho}{\rho^k}. \tag{24}$$

For $|s| \le B < \rho$,

$$|g(s) - q_P(s)| \le \sum_{k=P+1}^{\infty} |a_k|\,|s|^k \tag{25}$$

$$\le M_\rho \sum_{k=P+1}^{\infty} (B/\rho)^k = M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}. \tag{26}$$

Applying Lemma 1 and taking $\mathcal{S} = \mathcal{S}^\star$, $\beta_k = a_k$ for $0 \le k \le P$, yields the stated unnormalized bound. Dividing by $\sqrt{M}$, $M = \prod_{n=1}^{N} I_n$, gives the normalized form used in Theorem 3. $\square$

B.2 Rank Expansion

Proof of Proposition 1. Write the Tucker tensor entrywise as

$$\mathcal{S}_{i_1,\dots,i_N} = \sum_{r_1=1}^{R_1} \cdots \sum_{r_N=1}^{R_N} \mathcal{X}_{r_1,\dots,r_N} \prod_{n=1}^{N} A^{(n)}_{i_n,r_n}. \tag{27}$$

Taking an element-wise p-th power gives

$$(\mathcal{S}_{i_1,\dots,i_N})^p = \prod_{q=1}^{p} \left(\sum_{r_1^{(q)}=1}^{R_1} \cdots \sum_{r_N^{(q)}=1}^{R_N} \mathcal{X}_{r_1^{(q)},\dots,r_N^{(q)}} \prod_{n=1}^{N} A^{(n)}_{i_n,r_n^{(q)}} \right) \tag{28}$$

$$= \sum_{\gamma_1\in[R_1]^p} \cdots \sum_{\gamma_N\in[R_N]^p} \left(\prod_{q=1}^{p} \mathcal{X}_{r_1^{(q)},\dots,r_N^{(q)}} \right) \prod_{n=1}^{N} \left(\prod_{q=1}^{p} A^{(n)}_{i_n,r_n^{(q)}} \right), \tag{29}$$

where $\gamma_n = (r_n^{(1)}, \dots, r_n^{(p)})$. For each mode n, let $\alpha_n \in \mathbb{N}^{R_n}$ record the counts of the entries in γ_n, so $|\alpha_n| = p$. Then

$$\prod_{q=1}^{p} A^{(n)}_{i_n,r_n^{(q)}} = \prod_{r=1}^{R_n} \left(A^{(n)}_{i_n,r}\right)^{\alpha_{n,r}} = A^{(n)\langle p\rangle}_{i_n,\alpha_n}. \tag{30}$$

Group all ordered tuples $(\gamma_1, \dots, \gamma_N)$ that produce the same multi-indices $(\alpha_1, \dots, \alpha_N)$, and define the aggregated core

$$\mathcal{X}^{\langle p \rangle}_{\alpha_1,\dots,\alpha_N} = \sum_{\substack{\gamma_1,\dots,\gamma_N: \\ \mathrm{count}(\gamma_n)=\alpha_n \ \forall n}} \prod_{q=1}^{p} \mathcal{X}_{r_1^{(q)},\dots,r_N^{(q)}}. \tag{31}$$

The preceding display is exactly the Tucker reconstruction

$$\mathcal{S}^{\odot p} = [\![\mathcal{X}^{\langle p \rangle}; \mathbf{A}^{(1)\langle p \rangle}, \dots, \mathbf{A}^{(N)\langle p \rangle}]\!]. \tag{32}$$

The n-th Veronese factor has $\binom{R_n+p-1}{p}$ columns, giving the claimed Tucker rank bound. $\square$

B.3 Separation

Lemma 2 (Full-rank Veronese evaluations)**.** *Let $D = \binom{R+p-1}{p}$. There exists a matrix $\mathbf{A} \in \mathbb{R}^{D\times R}$ whose Veronese feature matrix $\mathbf{A}^{\langle p \rangle} \in \mathbb{R}^{D\times D}$ has rank D. Consequently, for $I \geq D$, a generic matrix $\mathbf{A} \in \mathbb{R}^{I\times R}$ has $\mathrm{rank}(\mathbf{A}^{\langle p \rangle}) = D$.*

Proof. The homogeneous degree-p monomials in R variables are linearly independent polynomials. Hence there exist evaluation points $x_1, \dots, x_D \in \mathbb{R}^R$ such that the matrix $(x_i^{\alpha})_{i,\alpha}$, $|\alpha| = p$, is nonsingular; otherwise every $D \times D$ evaluation determinant would vanish, which would imply a nontrivial linear dependence among the monomials. Taking these points as the rows of $\mathbf{A}$ gives the first claim.

For $I \geq D$, choose any D rows. The determinant of the corresponding $D \times D$ Veronese submatrix is a polynomial in the entries of $\mathbf{A}$ that is not identically zero by the first claim. Its nonvanishing is therefore a generic property, so $\mathbf{A}^{\langle p \rangle}$ has full column rank for generic $\mathbf{A}$. $\square$

Proof of Theorem 1. Let $D = \binom{R+p-1}{p}$. Choose generic factor matrices $\mathbf{A}^{(n)} = [a_1^{(n)}, \dots, a_R^{(n)}] \in \mathbb{R}^{I_n \times R}$, and define the CP-diagonal Tucker backbone

$$\mathcal{S} = \sum_{r=1}^{R} a_r^{(1)} \otimes \cdots \otimes a_r^{(N)}. \tag{33}$$

Its mode-n unfolding is

$$\mathcal{S}_{(n)} = \mathbf{A}^{(n)} \left(\mathbf{A}^{(N)} \odot \cdots \odot \mathbf{A}^{(n+1)} \odot \mathbf{A}^{(n-1)} \odot \cdots \odot \mathbf{A}^{(1)}\right)^{\top}. \tag{34}$$

For generic $\mathbf{A}^{(n)}$, $\mathbf{A}^{(n)}$ has rank R, and the Khatri-Rao product on the right has rank R. Thus $\mathrm{rank}(\mathcal{S}_{(n)}) = R$ for every n, so $\mathcal{S}$ has exact Tucker rank $(R, \dots, R)$.

Set $\beta_p = 1$ and $\beta_q = 0$ for $q \neq p$. Then $\mathcal{Y} = \mathcal{S}^{\odot p}$ is a valid degree-p NTD-PL tensor. Entrywise,

$$\mathcal{Y}_{i_1,\dots,i_N} = \left(\sum_{r=1}^{R} \prod_{n=1}^{N} A^{(n)}_{i_n,r}\right)^p \tag{35}$$

$$= \sum_{|\alpha|=p} c_\alpha \prod_{n=1}^{N} \prod_{r=1}^{R} \left(A^{(n)}_{i_n,r}\right)^{\alpha_r}, \tag{36}$$

where $c_\alpha = p!/(\alpha_1! \cdots \alpha_R!) > 0$. Let $\mathbf{B}^{(n)} = \mathbf{A}^{(n)\langle p \rangle} \in \mathbb{R}^{I_n \times D}$. By Lemma 2, each $\mathbf{B}^{(n)}$ has full column rank D for generic $\mathbf{A}^{(n)}$. The mode-n unfolding is

$$\mathcal{Y}_{(n)} = \mathbf{B}^{(n)} \,\mathrm{diag}(c) \left(\mathbf{B}^{(N)} \odot \cdots \odot \mathbf{B}^{(n+1)} \odot \mathbf{B}^{(n-1)} \odot \cdots \odot \mathbf{B}^{(1)}\right)^{\top}. \tag{37}$$

The diagonal matrix $\mathrm{diag}(c)$ is nonsingular. Since a Khatri-Rao product of full-column-rank matrices is full column rank, the right factor has rank D. Therefore $\mathrm{rank}(\mathcal{Y}_{(n)}) = D$. This holds for every mode n, and the exact Tucker rank of $\mathcal{Y}$ is $(D, \dots, D)$. $\square$

B.4 Masked Generalization

Theorem 4 (Masked-entry uniform convergence)**.** *Let $\Theta \subset \mathbb{R}^d$ be a bounded NTD-PL parameter set with*

$$d = \prod_{n=1}^{N} R_n + \sum_{n=1}^{N} I_n R_n + (P+1).$$

For $\theta \in \Theta$, define the full-entry risk

$$\mathcal{L}(\theta) = \frac{1}{M} \sum_{\mathbf{i}} \ell_{\mathbf{i}}(\theta), \qquad M = \prod_{n=1}^{N} I_n,$$

and the masked empirical risk

$$\widehat{\mathcal{L}}_m(\theta) = \frac{1}{m} \sum_{t=1}^{m} \ell_{\mathbf{i}_t}(\theta),$$

where $\mathbf{i}_t$ are sampled uniformly from tensor entries. Assume $0 \le \ell_{\mathbf{i}}(\theta) \le B_\ell$ and $\ell_{\mathbf{i}}(\theta)$ is G-Lipschitz in θ, uniformly over $\mathbf{i}$. If Θ is contained in a Euclidean ball of radius B_Θ, then with probability at least $1-\delta$,

$$\sup_{\theta \in \Theta} \left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \le 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}$$

for every $0 < \varepsilon \le B_\Theta$.

Proof of Theorem 4. Fix $0 < \varepsilon \le B_\Theta$, and let $\mathcal{N}_\varepsilon$ be an ε-net of Θ in Euclidean norm. Since Θ lies in a d-dimensional ball of radius B_Θ, it admits a net with

$$|\mathcal{N}_\varepsilon| \le \left(\frac{3B_\Theta}{\varepsilon} \right)^d. \tag{38}$$

For any fixed $\theta_0 \in \mathcal{N}_\varepsilon$, the random variables $\ell_{\mathbf{i}_t}(\theta_0)$ are independent and bounded in $[0, B_\ell]$. Hoeffding's inequality gives

$$\Pr\left(\left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \ge u \right) \le 2 \exp\left(-\frac{2mu^2}{B_\ell^2} \right). \tag{39}$$

Taking a union bound over $\mathcal{N}_\varepsilon$, with probability at least $1-\delta$,

$$\max_{\theta_0 \in \mathcal{N}_\varepsilon} \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \le B_\ell \sqrt{\frac{\log(2|\mathcal{N}_\varepsilon|/\delta)}{2m}}. \tag{40}$$

Substituting the net-size bound yields the stated stochastic term.

Now take any $\theta \in \Theta$, and choose $\theta_0 \in \mathcal{N}_\varepsilon$ with $\|\theta - \theta_0\|_2 \le \varepsilon$. By the uniform Lipschitz assumption,

$$|\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| \le G\varepsilon, \qquad |\widehat{\mathcal{L}}_m(\theta) - \widehat{\mathcal{L}}_m(\theta_0)| \le G\varepsilon. \tag{41}$$

Therefore

$$\left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \le |\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| + \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \tag{42}$$

$$+ |\widehat{\mathcal{L}}_m(\theta_0) - \widehat{\mathcal{L}}_m(\theta)| \tag{43}$$

$$\le 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}. \tag{44}$$

Taking the supremum over $\theta \in \Theta$ completes the proof. □

Remark 1. *The Lipschitz and bounded-loss assumptions are standard for compact parametric classes. For NTD-PL, they follow when the observed values, core, factor matrices, and polynomial coefficients are bounded and the polynomial degree P is finite. The resulting Lipschitz constant may depend polynomially on these bounds and on P, but the covering dimension remains the number of tied NTD-PL parameters rather than the number of parameters in an untied high-order Tucker expansion.*

C Additional Experiments

C.1 Controlled Synthetic Mechanism Checks

The controlled experiments in the main text are designed to isolate the role of the scalar link. Table 4 first checks a negative control: when the source is linear Tucker, NTD-PL does not improve by inventing nonlinear terms. The only exception is an affine offset, where the constant link term is the intended correction.

For the nonlinear controlled tensors, the nonlinearity is introduced as an orthogonal residual rather than by directly setting $Y = g(S^\star)$. Let $S^\star$ be the linear low-rank Tucker backbone and let g be one of the candidate scalar responses. We form

$$R^\star = g(S^\star) - \frac{\langle g(S^\star), S^\star\rangle}{\|S^\star\|_F^2} S^\star, \tag{45}$$

and, when $\|R^\star\|_F > 0$, normalize it as

$$\widetilde{R}^\star = \frac{\|S^\star\|_F}{\|R^\star\|_F} R^\star. \tag{46}$$

The observed tensor is then

$$Y_\alpha = \sqrt{1-\alpha}\, S^\star + \sqrt{\alpha}\, \widetilde{R}^\star, \qquad 0 \le \alpha \le 1. \tag{47}$$

By construction, $\langle S^\star, \widetilde{R}^\star\rangle = 0$ and $\|\widetilde{R}^\star\|_F = \|S^\star\|_F$, so α is exactly the energy fraction assigned to the nonlinear residual. This keeps the total energy fixed while moving the observation away from the linear Tucker backbone. The candidate residual families are $g(s) = s^2 + s^3$, $g(s) = (\exp(\kappa s) - 1)/\kappa$, and $g(s) = \tanh(\kappa s)$, with κ set from the input range.

The main text reports the nonlinear alpha sweep and degree sweep. Figure 6 shows that degree continuation activates higher powers without destabilizing the fit. Table 5 complements those figures with a small but consistent term-mass concentration effect: the dominant contribution usually lands on the degree that matches the injected response, while smooth odd links spread mass across nearby odd orders.

Table 4: Linear consistency on synthetic Tucker tensors. The $p > 1$ columns report effective higher-order NTD-PL contribution.

Method	bias = 0		bias = 0.5	
	RMSE↓	$p > 1$	RMSE↓	$p > 1$
Tucker	**0.0114**	–	0.0647	–
NTD-PL, $\beta_0 = 0$	0.0116	4.19%	0.0522	39.17%
NTD-PL, β_0 free	0.0117	3.19%	**0.0118**	2.98%

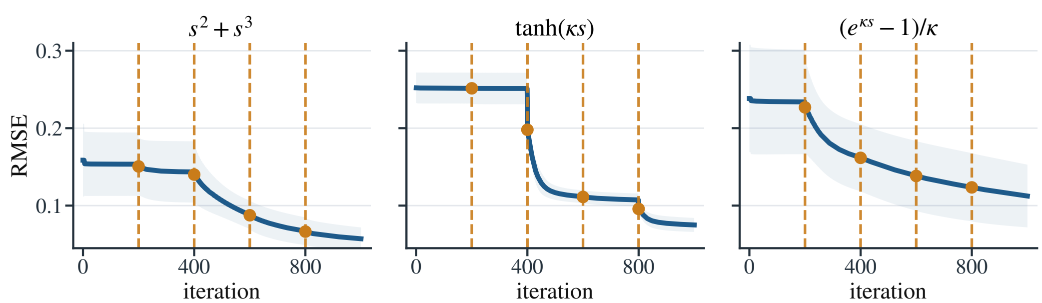

Figure 6: NTD-PL training curves for the controlled nonlinear mechanism checks at $\alpha = 0.25$ and $P = 5$. Dashed vertical lines mark degree-activation events.

Table 5: Representative effective NTD-PL term contributions under moderate nonlinearity, reported as percentages over ten seeds. Polynomial targets concentrate mass at the matching nonlinear degrees, while odd smooth links place most nonlinear mass on odd powers.

Link	Effective contribution (%)						
	$p=0$	$p=1$	$p=2$	$p=3$	$p=4$	$p=5$	$p=6$
s^2	0.5	43.3	**23.2**	5.8	8.2	9.1	9.8
s^2+s^3	0.1	24.1	9.8	**38.1**	7.3	11.5	9.1
$\sin(\kappa s)$	0.1	37.7	0.6	**35.4**	2.8	19.8	3.6
$\tanh(\kappa s)$	0.2	37.9	1.3	**30.9**	5.4	17.3	7.0

C.2 CAVE Completion Significance and Robustness

The main text reports the full-resolution random-missing CAVE completion table at rank $(33, 33, 4)$. We do not repeat that table here; instead, Table 6 gives the paired sign and Wilcoxon tests for the same protocol.

Table 6: Significance tests for full-resolution CAVE completion at rank $(33, 33, 4)$. Metrics are computed on missing entries.

	RMSE				SAM			
ρ	W/L	Med. gain	Sign p	Wilcoxon p	W/L	Med. gain	Sign p	Wilcoxon p
0.1	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	14/1	2.9742	9.77×10^{-4}	8.54×10^{-4}
0.3	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	13/2	1.9194	0.007	0.004
0.5	14/1	0.0026	9.77×10^{-4}	1.22×10^{-4}	13/2	1.6566	0.007	0.012
0.7	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	12/3	1.3394	0.035	0.048

Table 7: Link Refresh diagnostic on CAVE completion. Low/mid/high gains report the mean NTD-PL RMSE* gain over Tucker within each D_{link} tercile.

ρ	Spearman ρ_s	Low-yield RMSE* gain	Mid-yield RMSE* gain	High-yield RMSE* gain	Scenes
0.1	0.829	0.75%	7.92%	12.63%	15
0.3	0.871	-1.13%	9.10%	13.12%	15
0.5	0.889	-1.09%	9.16%	13.18%	15
0.7	0.868	-0.62%	9.22%	13.07%	15

C.3 Link Refresh Residual Diagnostic

The diagnostic used in the main text reuses the Link Refresh step from Section 5 as a one-round residual probe. Let $\widehat{\mathcal{Y}}_{\mathrm{T}}$ denote the Tucker output and $\mathcal{Y}$ the target tensor. Starting from this fitted Tucker latent tensor, we update only the scalar polynomial coefficients by fitting

$$h_\gamma(x) = \sum_{p=0}^{4} \gamma_p x^p$$

by ridge regression on pairs $(\widehat{Y}_{\mathrm{T},\mathbf{i}}, Y_{\mathbf{i}})$. In full reconstruction, all entries are used for this fit. In completion, the fit uses only observed entries and is then evaluated on held-out entries. The fitted map is applied entrywise to $\widehat{\mathcal{Y}}_{\mathrm{T}}$. We summarize this cheap refresh by the Link Refresh diagnostic

$$D_{\text{link}} = 10\log_{10}\left(\frac{\|\mathcal{Y}_{\text{eval}} - \widehat{\mathcal{Y}}_{\mathrm{T}}\|_F^2}{\|\mathcal{Y}_{\text{eval}} - \widehat{\mathcal{Y}}_{\text{link}}\|_F^2}\right),$$

which measures, on a log scale, how much Tucker residual energy is removed on the task-specific evaluation entries. In other words, it measures how much shared scalar-link yield is already present before the full NTD-PL alternating solver starts reshaping the Tucker parameters.

C.4 Joint Link-Family Smoke Test

The main model uses a polynomial link because it gives a closed-form ridge coefficient update and a direct Veronese rank-lift interpretation. To check whether the improvement is tied to the polynomial basis itself, we ran a small joint link-family smoke test on three CAVE scenes at spatial resolution 256×256 and backbone rank $(18, 18, 3)$. The spline variant replaces the polynomial link by a one-dimensional piecewise-linear link with eight knots. Its knot values are updated by ridge regression on the current latent entries, while the Tucker backbone is updated jointly through the link derivative.

Table 8: Joint scalar-link family smoke test on CAVE reconstruction. The spline link is slightly more flexible but slower; NTD-PL is the analyzable polynomial member used in the main paper.

Method	Params	RMSE↓	SAM↓	Time↓
Tucker	10k	0.0542 ± 0.0401	24.47 ± 10.95	1.4 ± 0.1s
NTD-PL(P=4)	10k	0.0502 ± 0.0363	23.23 ± 10.85	10.7 ± 0.1s
Joint Spline Link(K=8)	10k	0.0497 ± 0.0362	20.25 ± 8.01	40.4 ± 0.4s

C.5 CAVE Beads Pseudo-RGB Visualization

Figure 7 visualizes the CAVE beads scene under the reconstruction and random-completion protocols. Each 31-band tensor is rendered as pseudo-RGB by selecting three representative spectral bands near 75%, 50%, and 20% of the wavelength index range and applying a shared linear intensity scaling within each task. For completion, unobserved entries in the observation panel are shown in gray.

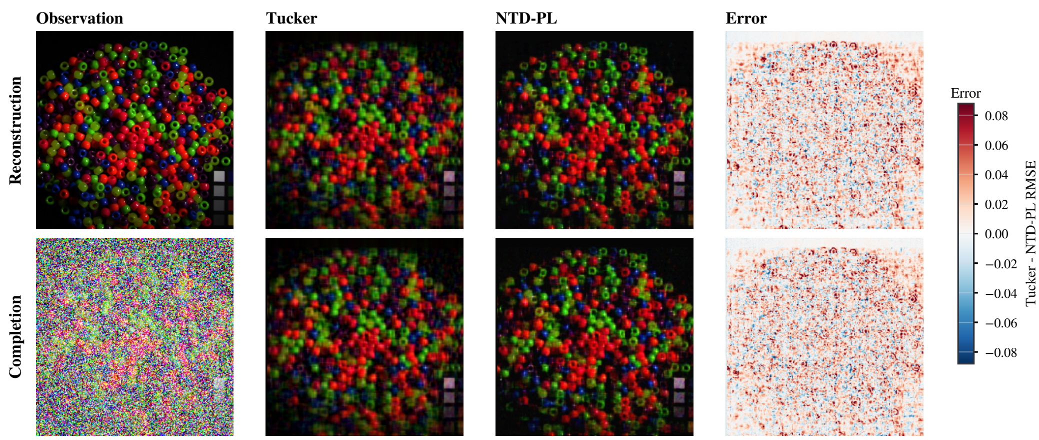

Figure 7: CAVE beads visual comparison. Rows correspond to reconstruction and random completion at missing rate 0.5. Error maps show Tucker minus NTD-PL per-pixel RMSE, so positive values indicate lower NTD-PL error.

C.6 Cross-Domain Real Tensor Checks

Table 9 reports matched-rank reconstruction on tensors outside the main hyperspectral setting: additional hyperspectral scenes, natural-image collections, and object-view tensors. The results are consistent with the scoped claim in Section 6.3: gains are clearest on hyperspectral and object-view data, while natural-image collections provide more modest improvements.

C.7 Non-HSI Link Refresh Diagnostics

The main diagnostic table uses per-unit non-HSI experiments rather than the collection-level tensors in Table 9. CBSD68 uses per-image units resized to $96 \times 96 \times 3$ with rank $(32, 32, 3)$; the available diagnostic CSV contains 16 CBSD68 units. CIFAR-10 uses 1000 class-balanced test images of shape $32 \times 32 \times 3$ with rank $(20, 20, 3)$. COIL-100 uses one object at a time as a $36 \times 48 \times 48 \times 3$ view tensor with rank $(8, 12, 12, 2)$. smallNORB uses one object instance at a time as an $18 \times 64 \times 64 \times 2$ stereo

Table 9: Additional matched-rank real tensor reconstruction. Columns report RMSE; gains are relative RMSE reductions over Tucker, and positive ΔRMSE favors NTD-PL.

Domain	Dataset	Shape	Rank	RMSE↓			Gain
				Tucker	NTD-PL	Δ	
Hyperspectral	Jasper Ridge	$100\times100\times198$	(11,11,43)	0.0462	0.0430	**0.0032**	**6.9%**
Hyperspectral	Samson	$95\times95\times156$	(10,10,35)	0.0263	0.0242	**0.0021**	**7.9%**
Hyperspectral	Urban	$307\times307\times162$	(43,43,47)	0.0444	0.0428	**0.0016**	**3.6%**
Natural images	CBSD68	$68\times96\times96\times3$	(20,32,32,3)	0.1327	0.1307	**0.0020**	**1.5%**
Natural images	CIFAR-10	$1000\times32\times32\times3$	(96,20,20,3)	0.0753	0.0724	**0.0029**	**3.8%**
Object views	COIL-100	$8\times36\times48\times48\times3$	(4,8,12,12,2)	0.0991	0.0931	**0.0060**	**6.0%**
Object views	smallNORB	$25\times18\times64\times64\times2$	(18,12,24,24,2)	0.0454	0.0429	**0.0024**	**5.4%**

view tensor with rank $(12, 24, 24, 2)$. All units are max-normalized independently, and diagnostic labels are assigned only from D_{link}, not from NTD-PL gain.

Table 10: Representative non-HSI units selected by D_{link}. Boundary/effective labels are assigned before inspecting NTD-PL gain.

Domain	Dataset	Unit	Label	D_{link}	Tucker RMSE	NTD-PL RMSE	Gain
Natural images	CIFAR-10	cifar10_test_06970	boundary	-0.136	0.0008	0.0008	-4.46%
Natural images	CBSD68	cbsd68_011	moderate	0.077	0.0409	0.0392	4.25%
Natural images	CBSD68	cbsd68_012	effective	0.341	0.0490	0.0408	16.69%
Object-view	COIL-100	coil_obj088	boundary	0.017	0.0280	0.0276	1.51%
Object-view	COIL-100	coil_obj091	effective	0.875	0.0751	0.0650	13.34%
Object-view	smallNORB	smallnorb_c3_i4	boundary	0.077	0.0315	0.0301	4.51%
Object-view	smallNORB	smallnorb_c2_i7	effective	1.048	0.0478	0.0380	20.53%

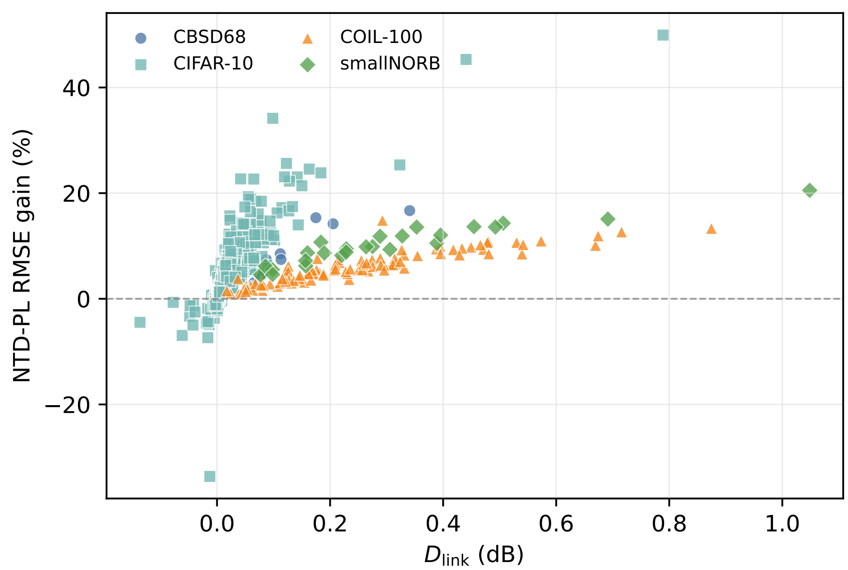

Figure 8: D_{link} versus NTD-PL RMSE gain over non-HSI tensor units. Units are selected independently and labels are assigned from D_{link}.

NeurIPS Paper Checklist

This draft uses a working checklist. It should be replaced by the official NeurIPS checklist distributed with the final 2026 submission template.

1. **Claims.** The main claims are stated in the abstract, introduction, theory section, and experiment captions: NTD-PL is a Tucker enhancement and shared-backbone rank lift for Tucker tensors; its

gains are tied to jointly reshaping the Tucker backbone under a shared scalar link rather than to modest rank increases alone.

2. **Limitations.** Section 7 discusses predicted near-linear boundary cases, whole-axis missingness, shared-link assumptions, and the current domain concentration in hyperspectral data.
3. **Theory assumptions.** The approximation results assume a bounded latent tensor and polynomial or analytic scalar links. The separation results use generic full-rank feature constructions. The masked generalization bound assumes bounded parameters, bounded observations, and random observed entries.
4. **Reproducibility.** The experiment section reports datasets, ranks, mask types, missing rates, and baseline definitions. The codebase includes scripts for rank sweeps, MLPCal, structured missingness, and the cross-domain real-tensor reconstruction study; the final version should add hardware details and exact command manifests.